

RHINO SERS

CATCH-PIT CHAMBER WITH SILT BUCKET

SERS300 | SERS450 | SERS600 | DATA SHEET



OVERVIEW

The RHINO SERS Series is engineered for the efficient catchment and removal of silt and debris from impermeable or hard surface areas such as paths, driveways, patios, roofs, car parks, roads and highways. A primary and secondary filtration system, with a removable silt bucket, gives effective primary treatment of surface runoff, improves downstream water quality and reduces the risk of flooding.

Each chamber is manufactured from high-density polyethylene using extrusion and thermoforming, and delivered fully assembled as a one-piece unit ready for installation, for light-duty to full-traffic loading applications.

SERS300
Ø300 mm

SERS450
Ø450 mm

SERS600
Ø600 mm

HOW THE SILT BUCKET WORKS

Surface water enters through the inlet and larger debris is intercepted as it drops into the chamber. The removable filter basket, or silt bucket, sits in the base below the pipe inverts and captures finer sediment before water leaves through the outlet. Maintenance is a lift-out and empty operation.

DESIGN COMPLIANCE

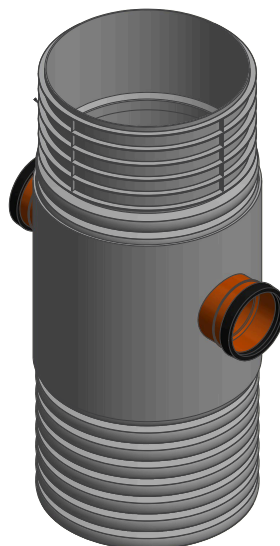
- EN 13598 Part 2 (2009) thermoplastic inspection chambers
- Design and Construction Guidance (DCG), Type D and E chambers
- Sewers for Adoption 7th Edition (SfA7)
- Building Regulations Part H1
- Adoptable and non-adoptable applications

CHAMBER OPTIONS

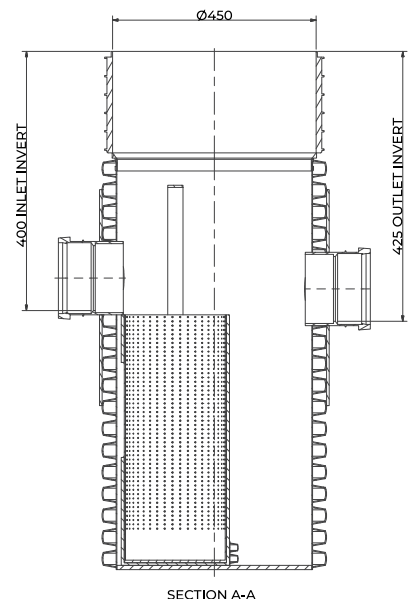
Pipework is drawn in 110 and 160 EN1401. Other pipework sizes are available subject to validation, and alternative inlet positions can be discussed with our technical team. Sump depths are standard or made to suit the project.

INSTALLATION

Suitable for applications that need above-ground inspection and maintenance of surface pipework. Loading class and bedding follow the installation guides; have the installation signed off by a suitably qualified professional.



SERS450 with manhole riser.



SERS450 section A-A. Dimensions in mm from drawing SERS450160-1.

NOT TO SCALE

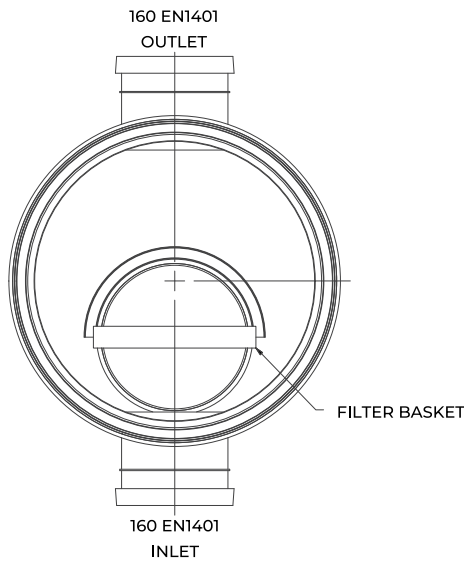
FEATURES & BENEFITS

- Primary and secondary filtration
- Removable silt bucket, lift out and empty
- Multiple inlet and outlet configurations
- Standard or bespoke sump depths
- Fits RHINO base pipe adaptors
- Anti-fracture thermoformed base
- Light-duty to full-traffic loading
- Chemical and corrosion resistant

SPECIFICATION

Sizes	SERS300, SERS450, SERS600 (Ø300, 450, 600 mm)
Material	HDPE, extruded and thermoformed
Filtration	Primary and secondary, removable silt bucket
Pipework	110 / 160 EN1401 as drawn, others on request
Sump	Standard or bespoke to project
Depth	Adoptable up to 2000 mm, non-adoptable up to 3000 mm to pipe soffit
Access	Max 350 mm opening where pipework is over 1 m deep

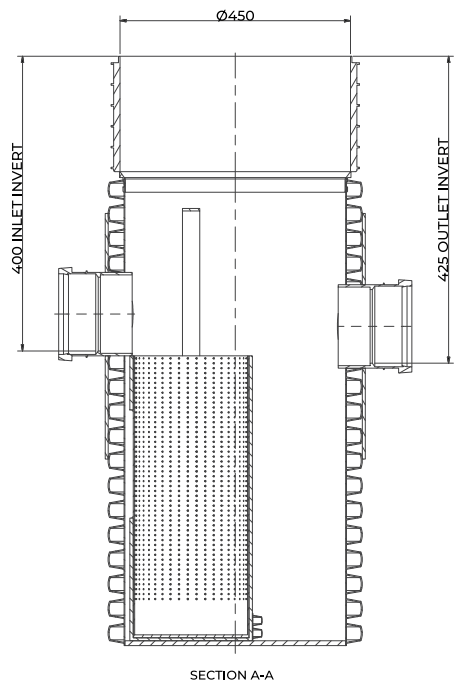
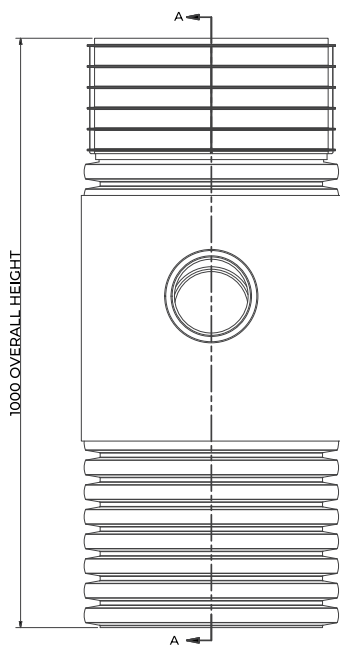
PLAN VIEW | OUTLET AT 12, INLET AT 6



SERS450160-1

Chamber	Ø450 mm
Overall height	1000 mm
Inlet	160 EN1401 at 6 o'clock
Outlet	160 EN1401 at 12 o'clock
Inlet invert	400 mm below top
Outlet invert	425 mm below top
Riser	1 x 450 manhole riser
Filtration	Removable filter basket

FRONT ELEVATION AND SECTION A-A



READING THE DRAWING

Drawing SERS450160-1 shows a Ø450 chamber with 160 mm EN1401 inlet and outlet and a 1 m overall height, topped with a 450 manhole riser. Invert levels are measured down from the top of the chamber.

The outlet invert sits 25 mm below the inlet invert. The filter basket occupies the base of the chamber, below both pipes.

Alternative inlet positions and pipework sizes are available subject to validation. Confirm the configuration with our technical team.

NOT TO SCALE Reproduced from the SuDS Enviro drawing pack for illustration. Dimensions in millimetres. Confirm project dimensions against the order drawing.

1

PRIMARY FILTRATION

Runoff from paths, roofs, car parks and highways enters through the inlet. Larger silt and debris are intercepted in the chamber rather than carried on into the downstream pipework.

2

SECONDARY FILTRATION

The removable silt bucket sits in the base below the inlet and outlet inverts and captures finer sediment. The dual system improves downstream water quality and reduces the risk of blockages and flooding.

3

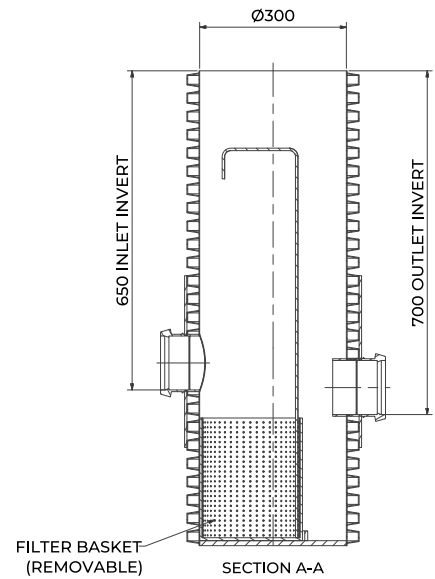
SIMPLE MAINTENANCE

The chamber is designed for above-ground inspection and maintenance. The silt bucket is lifted out, emptied and returned, keeping downstream drainage and attenuation clear.

4

ROBUST ONE-PIECE BUILD

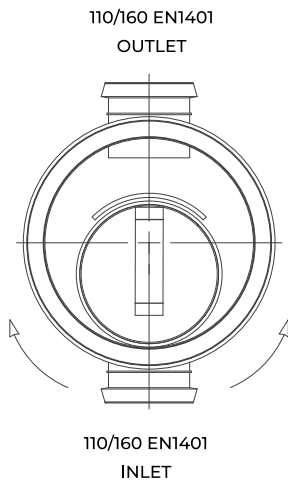
A thermoformed base gives structural strength and anti-fracture properties. Optimised internal surfaces help water flow and reduce blockages, and HDPE resists hydrogen sulphide, corrosive acids and bases.



SERS300 section A-A with removable filter basket. Drawing SERS300-1.

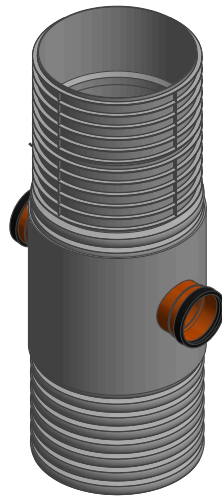
NOT TO SCALE

SERS300 | PLAN



Other pipework sizes available subject to validation. Alternative inlet positions available.

SERS600 | ISOMETRIC

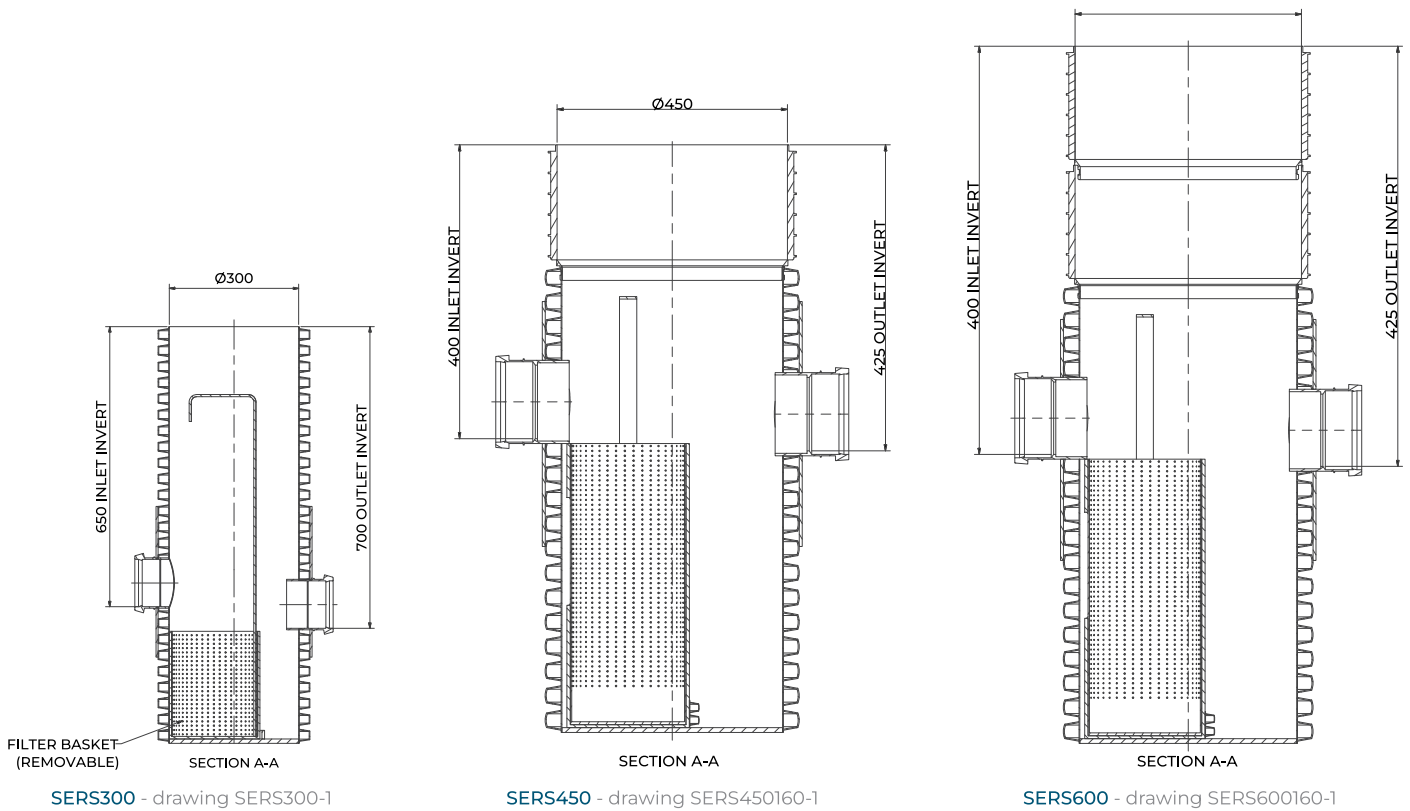


SERS600 with 2 x 600 manhole riser. Drawing SERS600160-1.

TO SPECIFY

- Chamber size: SERS300, SERS450 or SERS600
- Inlet and outlet pipe size and type
- Inlet clock positions
- Invert depths and cover level
- Sump depth, standard or bespoke
- Loading: light-duty or full-traffic
- Adoptable or non-adoptable

Our technical team confirms the configuration and returns an order drawing for approval.



NOT TO SCALE Sections are shown at different scales so each fits the column. Each is one drawn configuration; pipework, inverts and inlet positions are made to order.

CODE	CHAMBER	DRAWN PIPEWORK	OVERALL HEIGHT	INLET INVERT	OUTLET INVERT	RISER
SERS300	Ø300 mm	110 / 160 EN1401	1000 mm	650 mm	700 mm	-
SERS450	Ø450 mm	160 EN1401	1000 mm	400 mm	425 mm	1 x 450
SERS600	Ø600 mm	160 EN1401	1000 mm	400 mm	425 mm	2 x 600

Invert levels are measured down from the top of the chamber. All three sizes carry a removable filter basket (silt bucket) in the base.

INSTALLATION

- Bed and surround to suit the loading class and ground conditions
- Allow for variable groundwater levels
- Keep the chamber opening clear for lifting out the silt bucket
- Where pipework is over 1 m deep, fit restricted access of 350 mm maximum
- Have the installation signed off by a suitably qualified professional

COMPLIANCE

- EN 13598-2 (2009)
- DCG Type D and E chambers
- SfA7 (Sewers for Adoption)
- Building Regulations Part H1
- DCG restricted access: max 350 mm opening over 1 m depth

SPECIFY RHINO SERS

Tell us the chamber size, pipework and depths and our technical team will confirm the configuration.

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